

## EXPERIMENT-1

**Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.**

### AIM

To implement and demonstrate the FIND-S algorithm to find the most specific hypothesis based on a given set of training data samples.

### PROCEDURE

Import the required Python library.

Create the training dataset with attributes and target values.

Initialize the hypothesis with the most specific values.

Consider only the positive training examples.

Compare each positive example with the current hypothesis.

If an attribute value differs, replace it with ?.

Ignore negative training examples.

Display the final most specific hypothesis.

### CODE

```
import pandas as pd
data=[["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"], ["Sunny", "Warm", "High",
                                                                    "Strong", "Warm", "Same", "Yes"],
      ["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],
      ["Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes"]]
df=pd.DataFrame(data,columns=["Sky", "AirTemp", "Humidity", "Wind", "Water", "Forecast", "EnjoySport"])
print("Training Data")
print(df)
positive=df[df["EnjoySport"]=="Yes"].iloc[:,-1]
h=list(positive.iloc[0])
h=["?" if len(set(positive.iloc[:,i]))>1 else h[i] for i in range(len(h))]
print("\nMost Specific Hypothesis:")
print(h)
```

## OUTPUT:

```
Training Data
  Sky AirTemp Humidity   Wind Water Forecast EnjoySport
0 Sunny   Warm   Normal Strong  Warm   Same      Yes
1 Sunny   Warm    High  Strong  Warm   Same      Yes
2 Rainy   Cold    High  Strong  Warm  Change     No
3 Sunny   Warm    High  Strong  Cool  Change     Yes

Most Specific Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

## RESULT

The FIND-S algorithm was successfully implemented and demonstrated using Python in Google Colab. The most specific hypothesis obtained from the positive training examples is: